Unit 1 - DATA 607

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Set Options

APPROACH

Introduction

The data source is a simulated data set that I based on research we conducted examining the relationship between cognition and mobility gains in patient rehabilitation for stroke. For IRB and HIPPA reasons, I could not post the actual data set but the results and structure are similar though our original data includes over 1,000 variables.

The approach was to post the data on GitHub and then clean it in R Markdown as well as polish the names and value labels.

CODE

Import Data

Data were imported from GitHub as a CSV file.

```
url <- "https://raw.githubusercontent.com/cam5671/DATA-607/main/unit1/Mobility%20Gains%20in%20Stroke%20I
data <- read.csv(url)
head(data)
```

```
##   randomid AdmissionDate DischargeDate                               Notes age
## 1        1     4/13/2025      5/2/2025 Community ambulation goals achieved  86
## 2        2    12/15/2025    12/31/2025                                      74
## 3        3     9/28/2025    10/10/2025                                      78
## 4        4     4/17/2025     5/15/2025                                      87
## 5        5     3/13/2025     3/27/2025                                      90
## 6        6      7/8/2025     7/30/2025                                      66
##   sex los lestrength lestrengthdc cognition cognitiondc mobilityadm mobilitydc
## 1   0  19         29           36        22          23          15         46
## 2   1  16         29           43        23          24          15         51
## 3   0  12         78           92        25          26          41         84
## 4   0  28        101          102        18          19          17         21
## 5   0  14         74           88        19          20          34         78
## 6   1  22         67           78        22          24          22         55
```

Drop Unnecessary Columns

The purpose is to examine the effect of admissions factors (age, sex, and clinical variables) on the Mobility Score at discharge. The mobility score is part of section GG data for inpatient rehabilitation and collected on every patient across the US. The predictors include Cognitive Score as assessed by the Montreal Cognitive

Assessment, Lower Extremity Strength as measured by the Motricity Index, Age, Sex, and Length of Stay. Mobility score at discharge is the outcome. The most important predictor is likely the mobility score at admission which is also included. Discharge values for cognition and strength as well as hospital dates and clinic notes are unnecessary and are omitted.

```
data <- data[,-c(2:4, 9, 11)]
names(data)
```

```
## [1] "randomid"    "age"         "sex"         "los"         "lestrength"
## [6] "cognition"   "mobilityadm" "mobilitydc"
```

Change Variable Names

Variable names can be improved to be more presentable and legible. Changing names can be perilous in R if you use a vector of names and assignment them in order. I will change the names one at a time.

The final data set will include:

- ID: Participant ID
- Sex: Sex
- LOS: Length of Stay in Inpatient Rehabilitation
- Cognition: Cognitive Score on Montreal Cognitive Assessment
- MobilityADM: Mobility Score on Admission (Section GG of Inpatient Rehabilitation Assessment)
- MobilityDC: Mobility Score at Discharge

```
colnames(data)[1] <- 'ID'
colnames(data)[2] <- 'Age'
colnames(data)[3] <- 'Sex'
colnames(data)[4] <- 'LOS'
colnames(data)[5] <- 'LEStrength'
colnames(data)[6] <- 'Cognition'
colnames(data)[7] <- 'MobilityADM'
colnames(data)[8] <- 'MobilityDC'
head(data)
```

```
##   ID Age Sex LOS LEStrength Cognition MobilityADM MobilityDC
## 1  1  86   0  19         29        22          15         46
## 2  2  74   1  16         29        23          15         51
## 3  3  78   0  12         78        25          41         84
## 4  4  87   0  28        101        18          17         21
## 5  5  90   0  14         74        19          34         78
## 6  6  66   1  22         67        22          22         55
```

Value Labels

Sex is coded as 0 for Male, 1 for Female. The CSV file, however, did not include 'value labels' that we would typically see in STATA or SPSS or SAS. We can convert it in R to a factor variable.

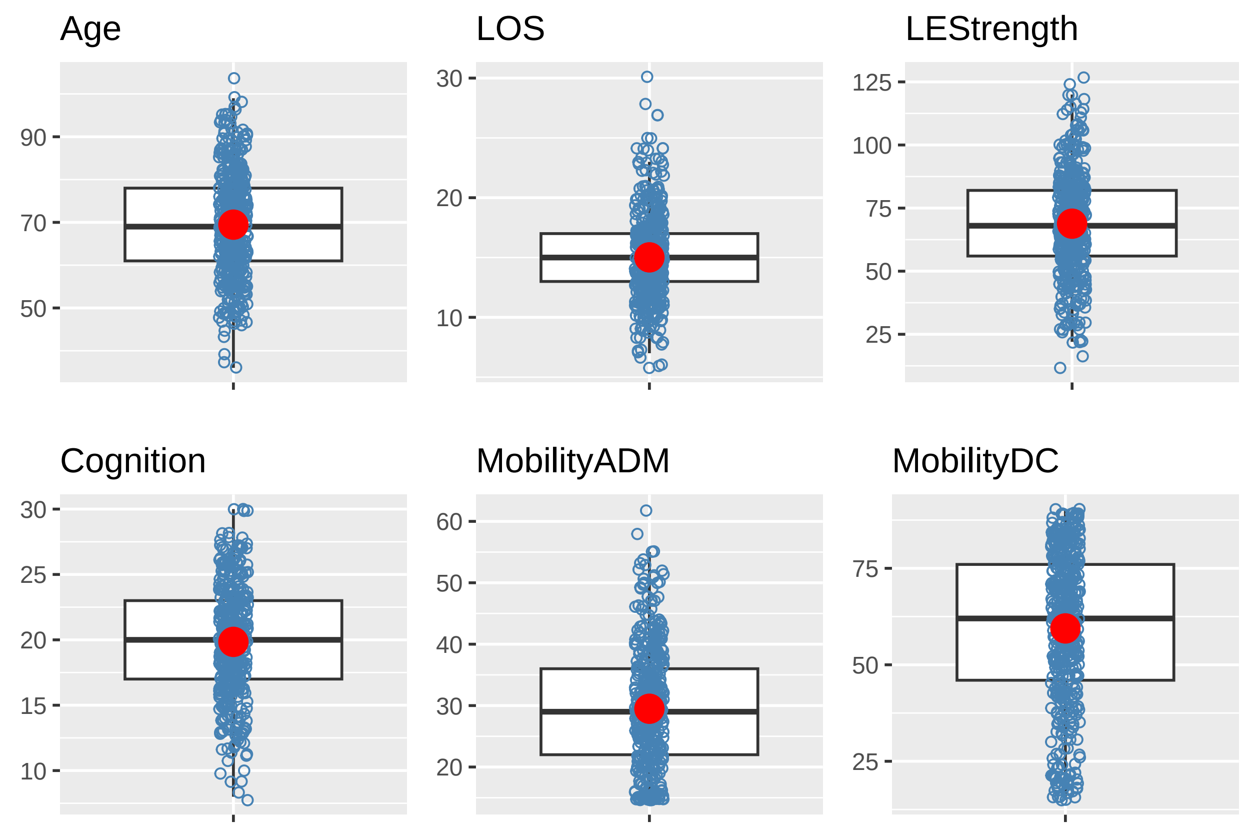
```
data$Sex <- factor(data$Sex, levels = 0:1, labels=c('Male', 'Female'))
table(data$Sex)
```

```
##
##   Male Female
##    264    189
```

Quick Screen of the Data

The data set is now established and clean. There are no missing values. The following is a quick screen of the data prior to what the main analysis was (linear regression though many other approaches are also possible). We already have a table for our categorical variable so plots for continuous variable will work here. Plots that show the data points, the medians and quartiles, as well as means provide the reader with a full picture of central tendency, skewness and outliers. The red dot is the mean superimposed on the boxplot. The variables are relatively normally distributed and you can see an increase in mobility from admission to discharge. When we simulate data using rnorm, the data end up artificially close normality. In our real data we see more skew with variables like Length of Stay (LOS).

```
library(ggplot2)
library(gridExtra)
library(ggplot2)
library(gridExtra)

plotf <- function(x) {

  ggplot(data, aes(x = "", y = .data[[x]])) +
    geom_boxplot(outlier.shape = NA) +
    geom_jitter(
      width = .05,
      color = "steelblue",
      pch = 1
    ) +
    stat_summary(
      fun = mean,
      geom = "point",
      pch = 16,
      color = "red",
      size = 5
    ) +
    labs(
      title = x,
      x = NULL,
      y = NULL
    )
}

vars <- c(
  "Age",
  "LOS",
  "LEStrength",
  "Cognition",
  "MobilityADM",
  "MobilityDC"
)

plots <- lapply(vars, plotf)

grid.arrange(
  grobs = plots,
  ncol = 3
)
```



Descriptive Statistics for Variables

The plots provide a full picture but descriptives can help as well.

```
library(psych)
library(gt)
library(dplyr)


vars <- c("Age", "LOS", "LEStrength", "Cognition",
          "MobilityADM", "MobilityDC")


desc <- psych::describe(data[vars])

desc_table <- desc %>%
  mutate(
    Variable = rownames(.),
    IQR = sapply(data[vars], IQR, na.rm = TRUE),
    Range = paste0(round(min, 1), "-", round(max, 1))
  ) %>%
  select(
    Variable,
    N = n,
    Mean = mean,
    SD = sd,
```

Descriptive Statistics

Variable	N	Mean	SD	Median	IQR	Range
Age	453	69.5	12.3	69.0	226.0	36–104
LOS	453	15.0	3.8	15.0	17.0	6–30
LEStrength	453	68.8	20.3	68.0	1.0	12–127
Cognition	453	19.8	4.1	20.0	4.0	8–30
MobilityADM	453	29.5	9.9	29.0	26.0	15–62
MobilityDC	453	59.4	20.0	62.0	6.0	15–90

```
    Median = median,
    IQR,
    Range
  )

desc_table %>%
  gt() %>%
  fmt_number(
    columns = c(Mean, SD, Median, IQR),
    decimals = 1
  ) %>%
  cols_align(
    align = "center",
    columns = everything()
  ) %>%
  cols_align(
    align = "left",
    columns = Variable
  ) %>%
  tab_header(
    title = "Descriptive Statistics"
  )
```

Conclusion

The data are ready for analysis and are appropriate for multiple linear regression though these data (and the real data) violate the assumption of heteroscedasticity and dominance analysis was used to determine the relative importance of the predictors. Other techniques like machine learning might offer better predictive accuracy and would be suitable for use on this dataset.

Acknowledgments

Generative AI (ChatGPT, OpenAI) was used for guidance on working with GitHub. It was also used to refine the loop used for the plots and it was used to format the table of descriptive statistics to improve the appearance from the standard output in the Psych package.